

CHAPTER 2

Emerging techniques for extraction and characterization of natural compounds

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2.1 Introduction

The bioactivity of plant extracts is closely related to their chemical composition. Therefore compounds isolated from plants present great potential to act as food preservatives and can benefit the consumer's health (Samtiya et al., 2021).

Recently the interest of the food industries in the use of plant extracts and essential oils (EOs) as natural food preservatives has considerably increased. Some examples of sources of these bioactive substances are garlic (Tavares et al., 2021), rosemary (Alavi et al., 2021), cloves (Haro-González et al., 2021), ginger ber (Beristain-Bauza et al., 2019), citrus fruits (Singh et al., 2021), coffee (Bondam et al., 2022), and hops (Arruda et al., 2021), among others.

Food industries often use conventional techniques to extract natural bioactive compounds, such as hydrodistillation, steam distillation, cold pressing, and solvent extraction (Zhang et al., 2018). However, nonconventional techniques, known as emerging techniques, have been studied to become viable by industries. Supercritical fluid extraction (SFE), ultrasonic-assisted extraction (UAE), microwave (MW)-assisted extraction (MAE), pulsed electric field (PEF)-assisted extraction, pressurized liquid extraction (PLE),

flash extraction, and deep eutectic solvents (DESs) are examples of these modern techniques that use green principles (Picot-Allain et al., 2021).

These emerging techniques present the following advantages: a greater extraction efficiency and selectivity, lower consumption of solvents and energy, possibility of operating at low temperature and pressure, and encouraging the use of environmentally friendly solvents. Furthermore, with the advancement of these green technologies, several studies have investigated agro-food coproducts as sources of natural substances to add value to the production chain and reduce industrial waste generation (Chemat et al., 2020).

In addition to using crude extracts, some compounds are isolated and used with a high degree of purity by the food industry. The most used techniques for the analysis of plant extracts containing bioactive compounds are gas chromatography (GC), classical liquid chromatography (LC), high-performance liquid chromatography (HPLC), two-dimensional liquid chromatography (HPLC-2D), high-speed countercurrent chromatography (HSCCC), supercritical fluid chromatography (SFC), and capillary electrophoresis (CE).

After isolation and purification the bioactive compounds can be analyzed and characterized by ultraviolet-visible spectroscopy (UV-Vis), Fourier transform infrared spectroscopy (FTIR), mass spectrometry (Ms), and nuclear magnetic resonance spectroscopy (NMR) (Altemimi et al., 2017).

2.2 Extraction of natural compounds

Natural substances may be extracted employing various extraction techniques. So far, conventional techniques are still mainstream in industry-scale applications. However, these widely used processes present some drawbacks that limit their application, such as time consumption, energy consumption, and high solvent consumption. Over the years, novel extraction techniques have been discovered and improved, enabling circumventing these limitations and, in addition, enhancing the extraction yield. Table 2.1 compares the long-term used conventional techniques and the emergent ones.

2.2.1 Conventional techniques

Although researchers have recently been interested in using plants as a promising green source of antimicrobials in the pharmaceutical and food

Table 2.1 A brief comparison between conventional and emergent extraction techniques.

Extraction technology	Principle	Solvent	Extraction parameters	Target compounds	Advantages (A)/limitations (L)	References
Conventional methods						
Hydrodistillation	Volatiles extracted from the plant matter in contact with boiling water	Water	High-temperature (boiling water)	EOs (nonpolar, low polarity)	(A) Low cost, simple operation, green technique/ (L) Low yield, time-consuming, degradation of thermolabile compounds	Quintana et al. (2021)
Steam distillation	Water vapor extracts volatiles from the plant material	Water/water vapor	High-temperature (boiling water)	EOs (nonpolar, low polarity)	(A) Low cost, simple operation, green technique/ (L) Low yield, time-consuming	Conde-Hernández et al. (2017)
Cold pressing	Mechanical process	Water or no solvent	Room temperature	EOs (eventually fatty acids) (nonpolar)	(A) Low cost, simple operation, green technique/ (L) Usually limited to citrus	Ou et al. (2015)
Solvent extraction	The solvent penetrates the plant matter and extracts the target compounds	Organic solvent		Phenolic acids, flavonoids, tannins (polarity is solvent-dependent)	(A) Simple operation, high yield/ (L) Environmental pollution, risks to human health, need for solvent removal by following methods	Gonelimali et al. (2018); Younus et al. (2021)
Emerging techniques						
Supercritical fluid extraction	Supercritical state of the solvent	Usually CO ₂	Room temperature	Nonpolar and low-molecular weight compounds	(A) Optimization of the extraction conditions, preservation of thermolabile compounds, green technique, high yield/(L) High cost	Conde-Hernández et al. (2017); Quintana et al. (2021)

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Table 2.1 (Continued)

Extraction technology	Principle	Solvent	Extraction parameters	Target compounds	Advantages (A)/limitations (L)	References
Ultrasonic-assisted extraction	The implosion of cavitation bubbles generates several effects that can damage the cell wall favoring greater energy and mass transfer across cell membranes.	Water, aqueous and nonaqueous solvents	Non-high-temperature or under heating/ atmospheric pressure/ short time	Thermolabile and unstable compounds/ bioactive, natural, and functional compounds/ phenolic compounds, anthocyanins/ vitamins, and others	(A) low cost; simple operating conditions; ease of control; ecologically friendly; low solvent consumption; temperature and time reduction/ (L) efficiency dependent on structural properties and liquid composition; production of free radicals; scalability and resistance of the equipment	Chemat, Rombaut, Meullemiestre, et al. (2017); Khadhraoui et al. (2021)
Microwave-assisted extraction	Nonionizing MW radiation induces molecular motion through ion migration and dipole rotation: transforming microwave energy into thermal energy. The heating by MW of water inside the treated biomaterial causes the evaporation of water molecules, inducing a high pressure inside the cell, resulting in its mechanical rupture.	Water, aqueous and nonaqueous solvents, and solvent free	Non-high-temperature/ atmospheric pressure/ short time	Resveratrol/ essential oils/ bioactive alkaloids/ antioxidant compounds/ vitamins and others	(A) requires less solvent; temperature and time reduction; easy operation and handling; low maintenance cost; higher extraction yields; lower energy consumption; reduced waste emissions/ (L) material needs to absorb microwave energy; heating uniformity	Alvi et al. (2022); Bener et al. (2022); Benmoussa et al. (2019); Chen (2013); Darvishzadeh & Orsat (2022); Khadhraoui et al. (2021); Khan et al. (2018); Patra et al. (2022); Sadeghi et al. (2017); Xiong et al. (2016)

Pulsed electric field-assisted extraction	PEF process is the application of short, repetitive high-voltage pulses to a material. The application of an electric field induces the formation of pores ("electroporation" mechanism) in cell membranes, increasing the membrane permeability and diffusion efficiency of the target compounds.	Water, aqueous and nonaqueous solvents	Non-high temperature or under heating/ atmospheric pressure/ short time	Raffinose/stachyose/ verbascose/betalains/ vulgaxanthine/phenolic compounds/ carotenoids/ anthocyanin and others	(A) operating at nonelevated temperatures; selective extraction; higher extraction yields; lower energy consumption; time reduction/ (L) difficulty in conductive materials; need to be combined with other techniques to increase yield	Andreou et al. (2021) ; Bouras et al. (2016) ; Boussetta et al. (2009) ; Bozinou et al. (2019) ; Hosseini et al. (2021) ; Khadhraoui et al. (2021) ; Martínez et al. (2018) ; Nowacka et al. (2019) ; Peiró et al. (2019) ; Segovia et al. (2015) ; Teh et al. (2015) ; Visockis et al. (2021)
Pressurized liquid extraction	Combined high-temperature and high-pressure extraction promotes an enhancement of the solubility of active compounds on solvents used; high temperature accelerates the extraction rate, and the high pressure prevents boiling at temperatures above the normal boiling point of the solvent.	Organic solvents (mainly ethanol) and/or water	High temperature and pressure/ short time	Depends on the solvent used, but mostly phenolic acids/ flavonoids/ anthocyanins/tanins/ prenylxanthohumol/ chlorophylls/ anthocyanins/gallic acid/p-hydroxybenzoic acid and others	(A) Possibility of using greener solvents; lower environmental impact; lower consumption of solvent and energy costs; shorter extraction times/ most of the instruments used are automated/ (L) Limitations when the samples are liquid/ deleterious effect on thermolabile bioactives depending on the temperature used (which may effect the extraction yield and the bioactivity of the extracts)	Alvarez-Rivera et al. (2019) ; Campanholi et al. (2020) ; Gil-Ramírez et al. (2012) ; Ho et al. (2022) ; Zandoná et al. (2020)

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Table 2.1 (Continued)

Extraction technology	Principle	Solvent	Extraction parameters	Target compounds	Advantages (A)/limitations (L)	References
Flash extraction	Ultra-high-speed shearing promotes the plant tissue damage and an ultrarapid dynamic molecular permeation, rapidly liberating the target compounds from plant cells; the extracted substances are quickly carried and infiltrated into the bulk solvent	Water and/or organic solvents; deep eutectic solvents	Usually performed at room temperature and pressure; short extraction times (1–5 min)	Polysaccharides, oils, proteins, phenols, alkaloids, terpenoids, and pigments, among others	Possibility of using green solvents; shorter extraction times; great extraction yields; allows the obtention of a wide range of natural compounds; lower solvent consumption; enables the extraction process to occur under mild conditions (room temperature and pressure); simple equipment requirement; possibility of working on industrial scale / (L) possibility of emulsification and blistering during the extraction, interfering on further filtration task; necessity of complex equipment for industrial scale; the solvent may interact with the metallic cutter reducing its service life	Chen et al. (2018) ; Liu et al. (2012) ; Qin & Xi (2021) ; Song et al. (2019)

Deep eutectic solvents	Similar to conventional solvent extraction, however, employing greener and more efficient solvents	Hydrogen bond acceptors (mostly based on choline chloride) + hydrogen bond donors (organic acids, carboxylic acids, polyalcohols, sugars, and urea)	Conditions depend on the extraction technique used (e.g., ultrasound, microwave, solid–liquid extraction)	Phenolic compounds (flavonoids, anthocyanins, phenolic acids, carotenoids), proteins, volatile monoterpenes, and lignocellulosic materials, among others	(A) thermal stability, polar variability, high boiling points, good dissolving and extractability, low toxicity or nontoxicity, especially concerning the solvents based on natural sources/ (L) further studies are needed concerning their toxicological aspects, needs a final step for the recovery of bioactives, high density	Hayyan et al. (2013) ; Hernández-Corroto et al. (2020) ; Ivanovi et al. (2020) ; Meng et al. (2018) ; Socas-Rodríguez et al. (2021)
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science areas, these bioactive substances have been used for different purposes by several cultures since ancient times (Elshafie & Camele, 2017; Sharifi-Rad et al., 2017). Classical methods widely known and used up to date for extracting the desired compounds from plants, such as hydrodistillation, steam distillation, cold pressing, maceration, and organic solvent extraction, are derived from historical methods documented in ancient manuscripts and books (Hamedí et al., 2013; Schlosser, 2011).

Hydrodistillation and steam distillation are the most commercially used techniques for obtaining EOs from several parts of plants, like leaves, flowers, seeds, and barks (DoTerra, 2022; Plant Therapy, n.d.). Their major advantages are their low cost, simple operation, and no need for organic solvents. Regarding hydrodistillation the plant parts are immersed in water inside an alembic or a Clevenger apparatus and heated. The vapor formed is a heterogeneous mixture of EOs and water, condensed back to a liquid phase, and collected. Since most EO compounds have little or no miscibility with water, their separation is achieved by decantation (Asbahani et al., 2015). Steam distillation is similar, except that the water and the plant material are placed in different recipients. In this case, only the water is heated, and the steam generated carries the volatile compounds. However, despite their simplicity, these methods present several downsides, such as a long extraction time, low yield, loss of polar molecules that remain in the water, and degradation of thermolabile compounds (Conde-Hernández et al., 2017; Quintana et al., 2021).

Another method commercially used is cold pressing or cold extraction. As the name implies, cold pressing is a nonthermal technique mainly used for extracting EOs from citrus' pericarp, and it consists of using a mechanical process to press, grind, and squeeze whole citrus fruits or their parts (as peels and/or seeds), sianosian (Li et al., 2014; Siano & Cautela, 2021). Besides the volatile compounds, cold-pressed EOs often present, in their composition, a minor nonvolatile fraction that consists of fatty acids, steroids, and other substances (Kapsaski-Kanelli et al., 2017).

Organic solvents can also assist the extraction of bioactive compounds such as flavonoids, phenolic acids, and tannins. Acetone, ethanol, and methanol, as well as their mixtures, are often described in the literature as the main organic solvents used to obtain plant extracts with several applications (antimicrobials, insecticidal, antioxidants, antiquorum sensing, among others) (Gonelimali et al., 2018; Takó et al., 2020; Younus et al., 2021; Yuan & Yuk, 2018). The choice of the solvent will have a direct impact on the success of the extraction and will depend on many factors

related to the plant, the bioactive compounds' features, and the nature of the solvent, which can be polar (ethanol, methanol), nonpolar (hexane, ether), or with intermediate polarity (Abubakar & Haque, 2020). This method, however, presents several drawbacks mainly involving the use of organic solvents: high costs, environmental pollution, risks to the handler's health, and the need for the following methods to ensure solvent removal from the obtained extract.

2.2.2 Emergent extraction techniques

The several disadvantages presented by the conventional extraction methods have propelled the search for new green technologies capable of extracting bioactive molecules from plants without using organic solvents and/or high temperatures. These novel and emergent techniques are discussed below.

2.2.2.1 Supercritical fluid extraction

SFE is a novel, ecofriendly technique used to obtain a range of bioactive compounds from plant materials through supercritical fluids. Due to several factors, carbon dioxide (CO_2) is extensively used as a solvent in this method, also regarded as CO_2 -SE; it is approved as a food-grade solvent, nonpolar, chemically inactive, cheap, and easily separable from the obtained extract (Marzlan et al., 2020; Uwineza & Waśkiewicz, 2020). However, it is worth noting that other solvents, such as water, ethanol, ethylene and propane, can also be employed in SFE, acting as solvents or cosolvents (Quintana et al., 2021).

This technique is based on the particular behavior of the supercritical fluids when specific conditions of pressure and temperature are reached, more specifically, when above their critical point. In this supercritical fluid region the fluid presents something between a gas-like and a liquid-like behavior with particular properties that will provide a higher dissolving power and, therefore, a greater extraction efficiency when compared to conventional techniques (Conde-Hernández et al., 2017; Quintana et al., 2021; Silva et al., 2021). The critical point of CO_2 , for example, is reached at 31°C and 7.38 MPa, being appropriate for the extraction of thermolabile nonpolar compounds. Another advantage of this method is optimizing the extraction parameters to obtain higher yields of certain compounds or enhance the extract's biological properties (Marzlan et al., 2020). However, the technique presents some drawbacks, mainly associated with the expensive cost of equipment.

2.2.2.2 *Ultrasound-assisted extraction*

Ultrasound (US) has been recognized as one of the most promising emerging technologies in terms of commercial interest due to its diverse applications as well as multiple advantages such as (1) low cost, (2) simple operating conditions, (3) ease of control, and (4) being ecologically friendly (Soares et al., 2019). This technology is based on the propagation of sound waves with a frequency above human hearing (> 20 kHz) and can be divided into high intensity–low frequency (intensity of $10\text{--}1000\text{ W/cm}^2$ and frequency of $20\text{--}100$ kHz) and low intensity–high frequency (intensity $< 1\text{ W/cm}^2$ and frequency > 1 MHz) (Arruda et al., 2021). Generally, high-intensity equipment is used for UAE, also known as ultrasonic extraction or sonication.

During sonication, sound waves propagate through the medium, causing a longitudinal displacement of the particles, which results in a series of alternating cycles of rarefaction and compression in the medium. These cycles can generate a phenomenon known as cavitation, which involves the formation, growth, and collapse of bubbles. Cavitation bubbles grow during rarefaction cycles and decrease during compression phases (Soares et al., 2020) (Fig. 2.1). The collapse of the bubbles produces a large amount of energy in a small area, which results in a high temperature (up to 4000°C) and pressure (up to 5000 atm) at specific points in short periods (milliseconds), which can generate turbulence and high shear force in the cavitation zone (Fig. 2.1) (Perera & Alzahrani, 2021; Wang et al., 2020).

In this context, during the UAE the implosion of cavitation bubbles generates microjets of extreme conditions of pressure and temperature that are capable of breaking up protein aggregates, causing unfolding of the protein structure, reducing the particle size, and breaking through the cell walls favoring greater energy and mass transfer across cell membranes (Fig. 2.1) (Khadhraoui et al., 2019; Zhang et al., 2018). This results in greater porosity and accelerated penetration of the solvent into the biological material (Chemat, Rombaut, Sicaire, et al., 2017; Khadhraoui et al., 2021), which contributes to accelerating the diffusion of compounds contained in the matrix (Kentish & Feng, 2014) (Fig. 2.1).

Another advantage of the UAE includes the low consumption of solvents and energy and the reduction of temperature and extraction time (Khadhraoui et al., 2019; Mason et al., 2011; Virot et al., 2010). Furthermore, this technology can be applied to extract thermolabile and

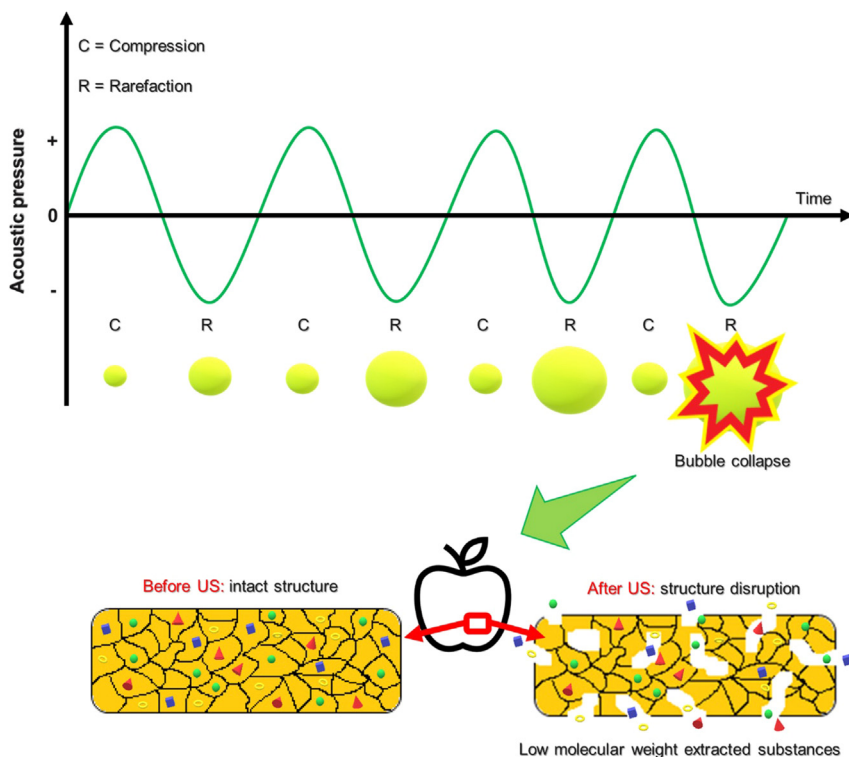


Figure 2.1 Illustrative schematic of the ultrasound extraction mechanism.

unstable compounds with higher yield and selectivity (Zhang et al., 2018). In this context, UAE is applied to extract several bioactive, natural, and functional compounds, including phenolic compounds, anthocyanins, and vitamins, among others (Barba et al., 2016; Chemat, Rombaut, Sicaire, et al., 2017). For example, Da Rocha and Noreña (2020) extracted bioactive compounds from grape pomace using ultrasound treatment (varying the power from 250 to 450 W for up to 15 min) and observed that extraction performed for 10 min resulted in maximum anthocyanin content.

However, the efficiency of the technology is closely related to the solid structural properties (Khadhraoui et al., 2019) and liquid composition (Chemat, Rombaut, Sicaire, et al., 2017) of the material (Khadhraoui et al., 2021). During processing, several physical mechanisms can occur (Chemat, Rombaut, Meullemiestre, et al., 2017), among the main ones: local erosion, shear forces, sonoporation, fragmentation, capillary effect,

and detexturation. These mechanisms can contribute to maximizing extraction yields by increasing the surface area of contact between water and product components (Khadhraoui et al., 2021). In this context, US can act through different mechanisms depending on the food matrix. Khadhraoui et al. (2019) evaluated the effect on blackcurrant, bitter orange, and artichoke leaves. Interestingly, for artichoke, considerable differences were noted between the upper and lower surfaces.

Thus it appears that US has a highly complex mode of action, and the extraction performance is directly correlated to the resistance level of the different matrices (Khadhraoui et al., 2021). In addition the level of degradation of the target compounds by the production of free radicals during sonication should be evaluated (Pingret et al., 2013). Another challenge is correlated with the scalability and resistance of the equipment, especially for probe ultrasound, which could make the use of this technology in industrial processes unfeasible. However, some advances in US technology have been circumventing these limitation points, aiming to expand its industrial application (Section 2.5).

2.2.2.3 Microwave-assisted extraction

MW technology has several food processing applications, including drying, pasteurization, sterilization, thawing, and extraction, among others (Khan et al., 2018; Sadeghi et al., 2017). In this context, MAE is one of the most used extraction methods because when compared to conventional extraction, it requires less solvent and is faster (Darvishzadeh & Orsat, 2022). In addition, MW heating is a technology of easy operation and handling and low maintenance cost (Cavalcanti et al., 2016). In contrast to convection or conduction heat transfer mechanisms, MW heating is a radiative heating mechanism that occurs with the interaction of electromagnetic radiation at certain frequencies with dielectric materials (Cavalcanti et al., 2016). During heating, MWs penetrate directly and quickly into the food, and this favors faster energy absorption by the treated material, reduced thermal gradients, and selective heating (Khadhraoui et al., 2021). Despite these advantages, one of the main challenges is correlated with heating uniformity, which depends on several intrinsic and extrinsic factors of the food (Cavalcanti et al., 2016).

MWs are electromagnetic waves composed of two perpendicular oscillating fields (electric and magnetic), whose spectrum frequency range varies from 300 MHz to 300 GHz, between radio and infrared waves (Cavalcanti et al., 2016). The two main mechanisms by which MWs

promote heat generation are (1) ionic conduction and (2) dipole rotation (Khadhraoui et al., 2021). Ionic conduction is due to the electrophoretic migration of ions when an oscillating electromagnetic field is applied. When ionic molecules are exposed to MWs the ions are forced to flow first in one direction and then in the opposite, according to the rapidly changing signal of the electromagnetic field. The resistance of the food molecules to this flow of ions as well as the collisions between the molecules caused by the continuous change in the direction of the ions results in friction by intermolecular collisions, promoting the volumetric heating of the food. In contrast, dipole rotation is related to the alternating motion of polar molecules that have a dipole moment (either permanent or induced by an electric field), which tries to align with the electric field. As the field decreases the clutter is restored, resulting in thermal energy release (Cavalcanti et al., 2016).

In many applications, both mechanisms occur simultaneously, effectively transforming MW energy into thermal energy (i.e., nonionizing MW radiation induces molecular motion through ion migration and dipole rotation) (Khadhraoui et al., 2021; Sadeghi et al., 2017). Consequently, there is a synergistic effect to accelerating extraction and improving extraction yield (Khadhraoui et al., 2021; Zhang et al., 2018).

In this context the heating by MW of water inside the treated biomaterial causes the evaporation of water molecules. Due to evaporation, high pressure is created inside the cell of the biomaterial, which forces the cell wall resulting in its mechanical rupture (Alvi et al., 2022). In parallel, solvent penetration into internal tissues is increased, and, therefore, higher extraction yields and selectivity can be achieved (Khadhraoui et al., 2021; Khan et al., 2018). Notably, there are two types of MAE methods: solvent extraction (generally for nonvolatile compounds) and solvent-free extraction (generally for volatile compounds) (Vinatoru et al., 2017; Zhang et al., 2018). In this case, when the water content of a food matrix acts as a solvent the process is termed solvent-free microwave extraction (SFME) as no solvent is added during the process (Alvi et al., 2022).

Key factors that affect MAE efficiency include the level and time of application of MW power, the temperature reached during the process, and the dielectric properties of the applied solvent and biomatrix (Darvishzadeh & Orsat, 2022). The main advantages of the MAE are ease of handling the equipment, reduction of the process duration, no need for solvent or moderate consumption (Chemat & Cravotto, 2012), selective and more efficient heating than conventional heating, higher extraction

yields, lower energy consumption, and reduced waste emissions (Khadhraoui et al., 2021; Sadeghi et al., 2017). However, the extraction solvent must absorb microwave energy (Khadhraoui et al., 2021). Fig. 2.2 shows an experimental setup of the MW extraction process.

Several works used MW technology to extract compounds of interest. Among them, the study by Chen (2013) optimized the MAE conditions to extract resveratrol. Benmoussa et al. (2019) used solvent-free MAE to extract EOs from seeds of *Foeniculum vulgare* Mill. Xiong et al. (2016) used MAE to extract five bioactive alkaloids: lyensinin, neferin, isoliensinin, dauricin, and nuciferin from lotus plumule. Darvishzadeh and Orsat (2022) successfully optimized the conditions of MAE of active antioxidant compounds in leaves and flowers of Russian olive. Patra et al. (2022) obtained a maximum extraction of 78.4 wt.% for sea buckthorn pomace extract using MAE at a temperature of 80°C and a power of 150 W with ethanol as a solvent (in Soxhlet extraction, the yield was recorded as only 44.5 wt.%). Furthermore, these authors found that MAE was more effective than conventional Soxhlet extraction to obtain antioxidant compounds with higher yields (Patra et al., 2022). In another study, Bener et al. (2022) used MAE from hazelnut pomace and flour samples to obtain antioxidant compounds. The results revealed that the MAE process for extracting antioxidants offers advantages such as short extraction times, low solvent requirements, multiple sample preparation, and ease of automation (Bener et al., 2022).

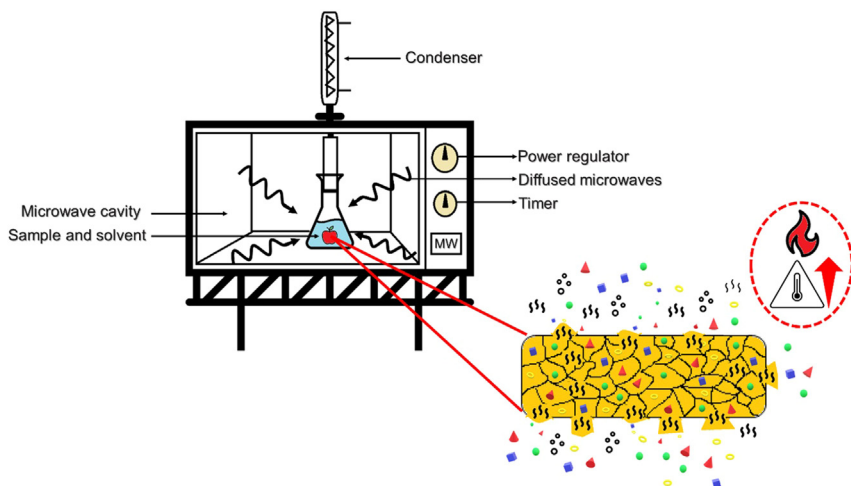


Figure 2.2 Illustrative schematic of the microwave extraction process.

2.2.2.4 Pulsed electric field-assisted extraction

The PEF process is the application of short (2–300 Ms), repetitive high-voltage pulses (20–80 kV/cm) to a material held between two electrodes (Nowacka et al., 2019). The efficiency of PEF extraction depends on the “electroporation” mechanism, also called “electropermeabilization” (Khadhraoui et al., 2021). The application of an electric field induces the formation of pores in cell membranes. Consequently the membrane permeability and diffusion efficiency of the target compounds are significantly increased (Bozinou et al., 2019; Khadhraoui et al., 2021).

In this context the pulsed electric field generates heat due to internal friction, and the accumulation of charged particles in the cell wall results in cell lysis due to electroporation. This could increase the extraction yield of several compounds (Lal et al., 2021) (Fig. 2.3).

The efficiency of PEF processing depends on several parameters, including electric field strength, processing time, specific energy, pulse number, pulse width and shape, frequency, and temperature. In addition, other product-related parameters are also influenced, such as cell size and shape, orientation in the electric field, as well as medium properties

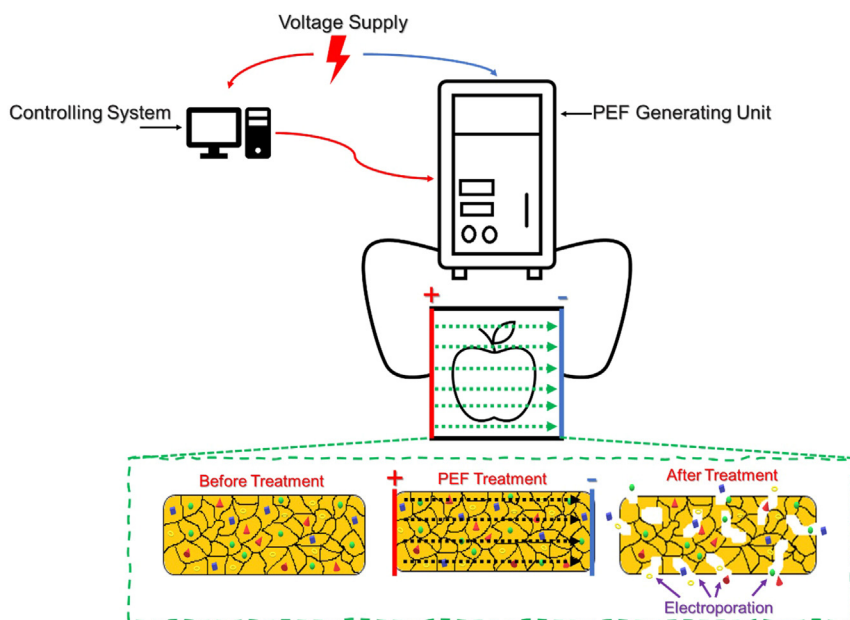


Figure 2.3 Illustrative schematic of the PEF extraction process.

(conductivity, composition) (Lal et al., 2021; Moens et al., 2020; Puértolas et al., 2012) (Fig. 2.3).

This technology can operate at nonelevated temperatures, thus preserving thermolabile compounds, allowing a selective extraction, and preserving color, flavor, and vitamin C content, among other food compounds (Khadhraoui et al., 2021). In addition the processing time can be reduced due to greater mass transfer increasing extraction yield (Bozinou et al., 2019).

Several studies in the literature demonstrate the positive impact of PEF in extracting compounds of interest. Examples include the increase in the extraction of (1) raffinose, stachyose, and verbascose from chickpea (Andreou et al., 2021); (2) polyphenols from the defatted canola seed cake (Teh et al., 2015); (3) betalains and vulgaxanthine from beet root (Nowacka et al., 2019; Visockis et al., 2021); (4) total phenolic compounds from Borage (*Borage officinalis* L.) (Segovia et al., 2015), grape skin (Boussetta et al., 2009), corn meal (Hosseini et al., 2021), and lemon waste residue (Peiró et al., 2019); (5) carotenoids from *Rhodotorula glutinis* (Martínez et al., 2018); and (6) anthocyanin from blueberry (Zhou et al., 2015) and grapes (Delsart et al., 2012). In a study, Bouras et al. (2016) evaluated the extraction of antioxidant compounds extracted from Norway spruce bark and found a higher phenolic content (eightfold) and antioxidant activity (30-fold) after PEF treatment compared to untreated samples.

However, PEF has some shortcomings related to the difficulty of using it with conductive materials as well as the necessity to be combined with other techniques to recover acceptable yields of the interest compounds (Khadhraoui et al., 2021; Khan et al., 2018). Thus the combination of PEF and US has shown promising results. Here, different bioactive compounds (phenolics, flavonoids, tannins, anthocyanins, antioxidant activity, and volatile compounds) were successfully extracted from almond seeds using PEF coupled with the US treatment (Manzoor et al., 2019).

2.2.2.5 Pressurized liquid extraction

PLE is regarded as a novel extraction technique that employs solvents at high temperatures and pressures, always below their critical points. This way the solvent is maintained in the liquid state during the whole extraction procedure (Alvarez-Rivera et al., 2019). As a result, PLE emerged with great potential for the obtention of green compounds since it can be conducted with water and/or organic solvents, targeting a wide range of

bioactives (Sumere et al., 2018). Furthermore, depending on the solvent used (e.g., ethanol, water), PLE can be considered a “green” and environmentally friendly extraction technique due to its lower energy consumption, reduced emission of pollutant solvents, and improved efficiency (Corazza et al., 2018; Soh & Eckelman, 2016).

The instrumentation needed to carry out a PLE process is relatively simple, with many options commercially available. Basically, it comprehends a solvent reservoir, a pump, an oven containing the extraction cell, different valves and restrictors, and a collecting vial. The differences that can be found are concerned if the instrumentations act in a static or dynamic mode (Alvarez-Rivera et al., 2019).

The PLE is based on a liquid-phase process. Generally the samples are solid/semisolid, concerning the commercial equipment, which is not a problem if the initial material are plants. However, a further process will be needed concerning liquid samples, such as adding an absorbent/adsorbent. The conditions used on PLE commonly occur between 100°C and 200°C and between 5 and 15 MPa, allowing the solvent to be at the liquid phase during the extraction process (Alvarez-Rivera et al., 2019; Gil-Ramírez et al., 2012).

The PLE process has the principles of solvent extraction, employing high temperatures. However, during PLE the considerable high pressure and temperature used provide greater solubility of the target compounds than techniques carried out at room temperature and atmospheric pressure (Pavez et al., 2019). In fact the conditions applied to facilitate the breaking of bonds between the analyte and the matrix improve the ability of the solvent to dissolve the compounds of interest, increasing the extraction yields (Aliaño-González et al., 2020; Cejudo-Bastante et al., 2021). It is important to mention that the extraction efficiency depends on flow rate, pressure, temperature, and time used in the extraction, which will interfere with the matrix effect, mass transfer, and solubility of the bioactive compounds (Alvarez-Rivera et al., 2019).

Several studies have been published regarding the application of PLE for the obtention of natural compounds (Aliaño-González et al., 2020; Lama-Muñoz et al., 2020; Santos et al., 2021), including those from industry byproducts (Cejudo-Bastante et al., 2021; Katsinas et al., 2021).

Recently a study conducted by Campanholi et al. (2020) emphasized the great potential of employing PLE for the obtention of *Tetragonia tetragonoides* (Pall.) Kuntze extracts rich in antimicrobial compounds. PLE was applied using 80% ethanol as a solvent within the following conditions on

the extraction cell: 75°C and 80 bar. The extraction procedure lasted for 240 min. On the other hand the total time of extraction of the conventional extraction, using a 1:2 (vol/vol) petroleum/methanol ether mixture, at room temperature, was approximately 20 h. In addition to using cleaner solvents and presenting a lower extraction time the emergent technology enhanced the extraction yield, providing extracts rich in chlorophylls exhibiting great activity against *Staphylococcus aureus*-resistant strains.

One of the main characteristics of PLE, besides its greener profile, is the shorter extraction time (around 15 min) (Aliaño-González et al., 2020). Comparing a conventional extraction technique (Soxhlet) to PLE, Soxhlet extraction exhibited better performance in extracting phenolic compounds and mannitol from olive leaves (Lama-Muñoz et al., 2020). Nevertheless, the PLE was satisfactory in recovering bioactives from the plant matrix, presenting a considerably shorter extraction time (5 min vs 4 h), which contributed to a lower solvent and energy cost consumption. This can be an interesting advantage, especially concerning industrial applications.

Furthermore, PLE equipment allows inert extraction conditions (e.g., N₂ atmosphere) (Aliaño-González et al., 2020). The extraction of natural bioactives under inert conditions allows a higher recovery yield since when exposed to an O₂ atmosphere, especially at higher temperatures, the compounds are more prone to oxidative reactions, losing their activity.

PLE was also applied on *Pseuderanthemum palatiferum* (Nees) Radlk. leaves for obtention of bioactive compounds (Ho et al., 2022). The target substances were total phenolic compounds, especially gallic acid and *p*-hydroxybenzoic acid. The authors found that PLE (using water as a solvent) is greatly appropriated for the obtention of the mentioned bioactives. Furthermore the extracts exhibited interesting biological activity (i.e., antioxidant, antimicrobial, antihypertensive activities, and cytotoxicity against cancer and regular cell lines); the optimized PLE procedure (186°C, 8.8 mL/min, and 30% of aqueous ethanol) allowed the obtention of a high amount of other valuable components. The results indicated that PLE is a suitable technique for extracting natural biologically active compounds from *P. palatiferum* leaves.

In addition to the plant matrices mentioned, PLE has also been investigated for the obtention of naturally bioactive substances from other sources, such as propolis (Bankova et al., 2021), hops (Čulík et al., 2009;

Gil-Ramírez et al., 2012), açai (Aliaño-González et al., 2020), and goldenberry (Corazza et al., 2018), among others.

2.2.2.6 Flash extraction

First, flash extraction (also called “tissue-smashing extraction”) was applied as a pretreatment to extract tannins from herbs by a pharmacist in Japan. The results indicated an impressive amount of tannins with less samples and a short time (Okuda et al., 1989). Then, several years later, numerous investigations have been performed on both instrumentation and/or application of the flash extraction technique (Qin & Xi, 2021). The main characteristics of flash extraction are high yield in an extremely short time with simple operation and cheap equipment and better industrial acceptability (Cheng et al., 2016; Liu et al., 2012).

The instrumentation involved in flash extraction is called a flash extractor. This equipment is designed for rapidly smashing plant materials. In the literature, there are four types of flash extractors described: analytical, experimental, pilot, and industrial. The experimental equipment is mainly composed of a control system, electromotor, cutter head, extraction pot, and pedestal. On the other hand the industrial extractor is more complex, presenting a more elaborated structure; in addition to the experimental parts, some auxiliary facilities are added to adapt to the industrial production process (Liu, 2007; Qin & Xi, 2021).

The flash extraction occurs based on solvent extraction; however, its ultra-high-speed rotation of the specially designed cutter head (15,000–30,000 rpm) drastically crushes the plant material (c.a., 50–80 μm), promoting an enhanced exposure of the tissues (large transfer area) and, consequently, improving the extraction process (Fan et al., 2014). On the other hand the negative pressure caused by the cutter rotation causes a prominent disturbance, and once the bioactives are released from the plants' cells, they are rapidly carried out by the solvent (Fig. 2.4) (Fan et al., 2014; Yin et al., 2020). Therefore the target compounds for flash extraction will depend on the characteristics of the solvent used (e.g., polarity).

Since the equipment (cutter) is metallic, it is important to employ solvents with reasonable characteristics, especially noncorrosive profiles. Greener solvents with nontoxic, safe, low-cost, and recyclable aspects are excellent for flash extraction. Water and organic solvents, such as ethanol, methanol, acetone, *n*-hexane, and petroleum ether, are conventional solvents that are widely applied to flash extraction processes (Cheng et al.,

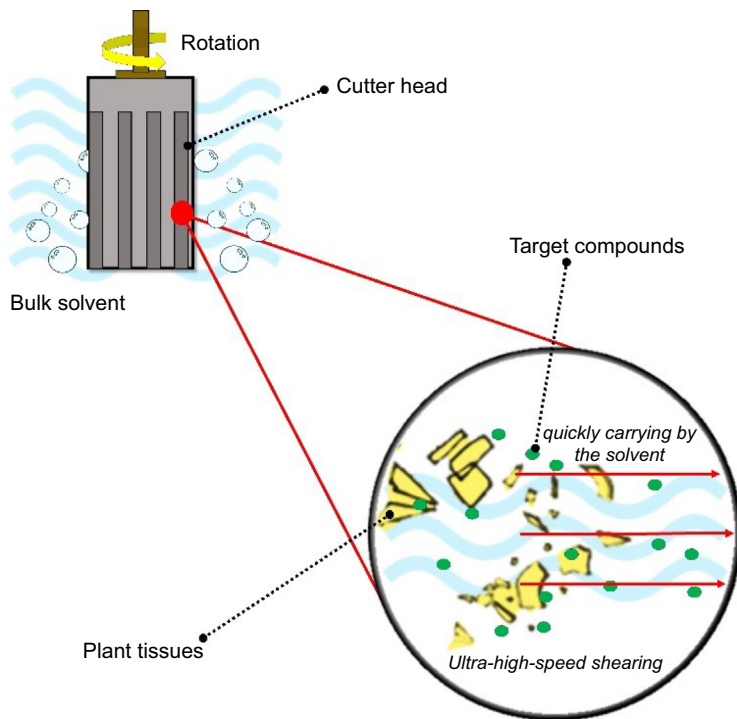


Figure 2.4 Representative scheme of the flash extraction mechanism.

2016; Qin & Xi, 2021). However, a limitation for extraction of water-soluble constituents is that when applying water as a solvent (a clean solvent), as the plant components are dissolved, the aqueous solution becomes viscous and prone to some processes (e.g., emulsification and blistering). This is due to the flash extractor's rapid shearing force and intensive stir, impairing the subsequent recovery treatment (Qin & Xi, 2021).

Among the main natural constituents obtained by flash extraction the following can be mentioned: polysaccharides, oils, proteins, phenols, alkaloids, terpenoids, and pigments, among others (Baggio, 2017; Chen et al., 2018; Song et al., 2019).

Exemplifying the flash extraction application, Liu et al. (2012) applied flash extraction to obtain limonin from orange (*Citrus reticulata* Blanco) seeds. The technique was optimized at 4000 rpm for 2 min, using 72% ethanol, at a 29:1 mL/g solvent:solid ratio. The equipment used was the Herbal Blitzkrieg Extractor (HBE), a typical flash extractor applied for

herbal constituents. Considering the relatively simple extraction conditions (room temperature and pressure) the short time guaranteed an impressive extraction of limonene (6.8 mg/g), achieving a purity of 95% and a recovery yield of 97.1%. Therefore flash extraction proved to be an efficient process for limonin extraction from orange seeds.

A study conducted by Yin et al. (2020) employed different natural deep eutectic solvents (discussed in Section 2.2.2.7) to obtain flavonoids of *Oroxylum indicum* seeds. First the flash extraction was carried out on a simple smashing tissue extractor (Ultra-Turrax), at room temperature, for 0.5–1.5 min. Then the optimum conditions were predicted using 5 mL of choline chloride/1,4-butanediol (1:3, 40% water content) for 1 min of extraction. The authors concluded that flash extraction not only improves the extraction yield but also saves a lot of time and energy compared to another emergent extraction technique (UAE) employing methanol as a solvent.

2.2.2.7 Deep eutectic solvents: a greener approach

DESs were introduced in 2004 (Abbott et al., 2004), emerging as a powerful green alternative to organic solvents and conventional ionic liquids. By definition, DESs are a mixture of two or more compounds that, under certain circumstances (i.e., under optimal conditions in terms of temperature and stirring time), present a melting point lower than that of either of its components (Hayyan et al., 2013). In addition, DES mixtures are liquid at room temperature or lower temperatures and have unusual solvent properties (Aroso et al., 2017). This characteristic is attributed to the generation of intermolecular hydrogen bonds between hydrogen bond donors and hydrogen bond acceptors presented on DES composition (Khandelwal et al., 2016; Meng et al., 2018).

The most common component used as a hydrogen bond acceptor is choline chloride (ChCl), a quaternary ammonium salt with a melting point of 302°C. Meanwhile, as donors, there are many substances, including organic acids, carboxylic acids, polyalcohols, sugars, and urea (Cui et al., 2017; Ivanović et al., 2018).

DESs are environmentally friendly, easily synthesized, biodegradable, nonvolatile, nontoxic, highly stable, low-cost solvents (Benvenuti et al., 2019) and can be applied in numerous extraction techniques, including the emergent ones described in the present chapter. These solvents' main properties present thermal stability, polar variability, high boiling points, good dissolving, and extractability and are easily eliminated

(biodegradability). When DESs are composed of natural eutectic compounds from plant metabolites or their derivatives, they are called natural deep eutectic solvents (NADESs). DES or NADES formation is usually conducted with the mixtures being heated under temperatures around 50°C–85°C, with continuous stirring until a homogeneous liquid appears (Socas-Rodríguez et al., 2021). A rotary evaporator instrument is commonly applied to DES preparation (Vázquez-González et al., 2020). Also, water may be added to reduce the mixture viscosity since higher densities and viscosities presented by DES/NADES (in comparison with conventional solvents) could be an issue at an industrial scale or in continuous-flow reactions (Ruesgas-Ramón et al., 2017).

As DESs are constituted by a mixture of components with distinct characteristics, their composition affects their properties and chemical characteristics, thus affecting their performance in the extraction of the target compounds (Wojeicchowski et al., 2021). Depending on the polarity and solubility of DESs, different compounds may be extracted. Among the main compounds extracted by DES, phenolic compounds (e.g., flavonoids, anthocyanins, phenolic acids, carotenoids), proteins, volatile monoterpenes, and lignocellulosic materials can be mentioned; these solvents can be applied as an extraction solvent or even as a delivery medium of the extract (Ivanovi et al., 2020; Murador et al., 2019; Ruesgas-Ramón et al., 2017).

Despite being relatively recent, various studies have been performed investigating the potential of DES and NADES on the extraction of natural bioactives from different sources: lemon verbena (*Lippia citriodora*) (Ivanović et al., 2018), pomegranate peel (*Punica granatum* L.) (Hernández-Corroto et al., 2020), strawberry and raspberry waste (Vázquez-González et al., 2020), rosemary (*Rosmarinus officinalis* L.) (Wojeicchowski et al., 2021), and safflower (*Carthamus tinctorius* L.) (Tong et al., 2021), among others.

For example a study aimed to evaluate choline chloride-based DES to extract phenolic compounds from rosemary leaves (Wojeicchowski et al., 2021). The authors found an optimum solid–liquid extraction (150 min) condition using choline chloride:1,2-propanediol at 65°C, a liquid:solid ratio of 40:1, and 50% (wt/wt) of water. The extracted fraction presented a great number of phenolic compounds, exhibiting a pronounced antioxidant and an interesting antimicrobial activity toward all the Gram-positive and Gram-negative bacteria tested (*Staphylococcus aureus*, *Escherichia coli*, *Clostridium perfringens*, *Listeria monocytogenes*, and *Salmonella* spp.).

Considering the assisting extraction potential of DES a study conducted by [Tong et al. \(2021\)](#) investigated 22 NADES consisting of food-grade ingredients screened in UAE (50°C, 200 W power for 30 min) of water-soluble chalcone derivates from safflower. Among the solvents tested, L-proline-acetamide was an efficient solvent for obtention of the plant bioactives, with higher extraction yields compared to water extraction. As the results indicated, NADESs present an excellent efficiency in the obtention of plant extracts and enable the enhancement of the bioactivity and bioavailability of the bioactive ingredients.

Concerning the products obtained using DES/NADES the recovery of the extracted bioactive substances from the structure of the solvent remains a significant challenge in producing “dried” extracts required for use in the various industrial sectors ([Ivanovi et al., 2020](#)). However, some studies have already demonstrated satisfactory results for bioactives recovering when using techniques such as macroporous resins ([Cui et al., 2018](#)) and antisolvents (e.g., water, ethanol) ([Tian et al., 2019](#)). It is also worth mentioning that regulatory limitations may still exist considering the DES/NADES employment. Since some of these solvents have not been appropriately registered, even though a few companies manufacture them and put them on the market, in some cases, the registry contains components of DES only.

Furthermore, another aspect related to DES/NADES concerns their toxicological aspects. An important characteristic of DES, especially NADES, is their low toxicity ([Socas-Rodríguez et al., 2021](#)). However, it was found that the cytotoxicity of a couple of choline chloride-based DESs was higher than that of their components (e.g., glycerin + ChCl), indicating that their toxicological behavior is different and the needing for further studies in the field ([Hayyan et al., 2013](#)). Nevertheless, this can be significantly reduced when NADESs are applied as solvents since it is allowed to use them in food and pharmaceutical formulations ([Benvenuti et al., 2019](#)).

2.3 Purification techniques

Many scientific papers report secondary metabolites from plants, which show food preservative properties ([Masyita et al., 2022](#); [Skroza et al., 2019](#); [Wen et al., 2021](#)). EO are sources of terpenes and/or phenylpropanoids, beyond phenolic compounds; moreover, flavonoids of different classes (such as flavanols, coumarins, and catechins) are an object of

growing interest in the food industry. The hydroxyl ($-\text{OH}$) moiety in the flavonoids, such as (+)-catechin (**1**) and rutin (**2**), and in phenylpropanoids, such as thymol (**3**) and carvacrol (**4**, Fig. 2.5), is directly related to the antimicrobial potential of these metabolites since the $-\text{OH}$ group can interact with the cell membrane of the bacteria causing its break and loss of cell components (Gyawali & Ibrahim, 2014).

Due to the need to obtain these substances with high purity to guarantee their bioactivity, even for evaluating the quality of commercial samples, which present these metabolites in their composition, more sophisticated compound separation methods, which are really effective and fast, have been used search. Thus chromatography is a good technique indicated for this purpose. Chromatographic techniques used to separate compounds from EOs and flavonoids, which help control food-contaminating microorganisms, will be described in sequence.

2.3.1 Gas chromatography

Gas chromatography is a traditional method used to separate organic compounds (de Jesus et al., 2020). It is a technique in which the gas phase analyte is transported through the column by a carrier gas (mobile phase), which can be He, N_2 , or H_2 (Harris, 2005) (Fig. 2.6). This separation technique may occur by gas–solid adsorption, where the analyte is adsorbed by the solid particles that constitute the stationary phase, or by gas–liquid partition, in which the stationary phase is a nonvolatile liquid (Skoog & Holler, 2014). Generally, gas chromatography is associated with a mass spectrometer (GC–Ms). It is possible to observe the fragmentation pattern of the compounds of a given sample and then establish their composition by comparing the mass

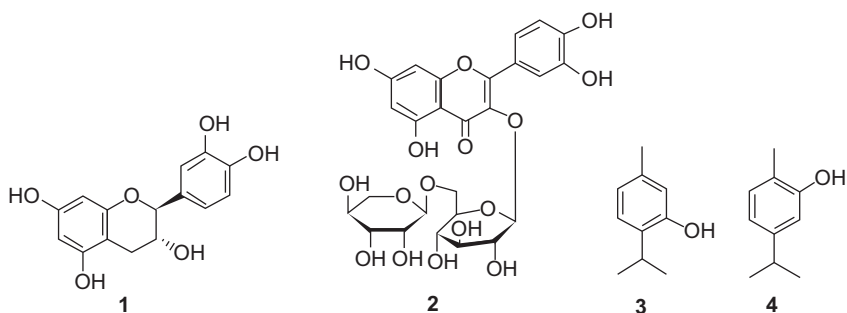


Figure 2.5 Chemical structures of compounds that exhibit antimicrobial activity: (+)-catechin (**1**), rutin (**2**), thymol (**3**) and carvacrol (**4**).

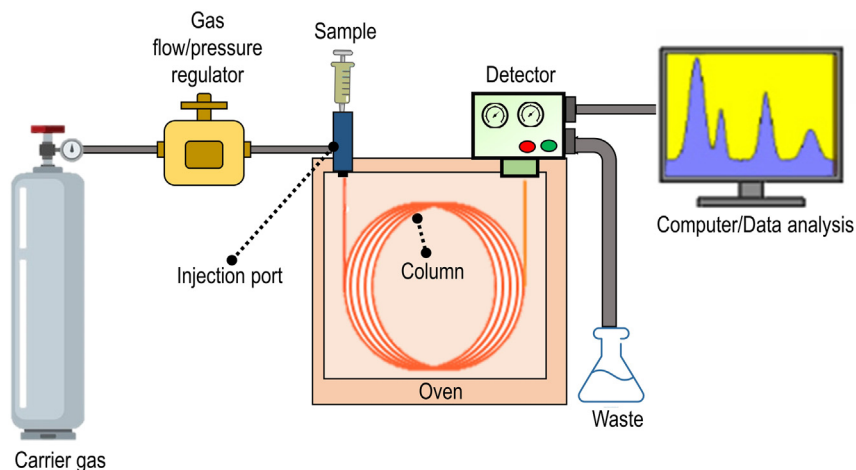


Figure 2.6 Scheme of a gas chromatography.

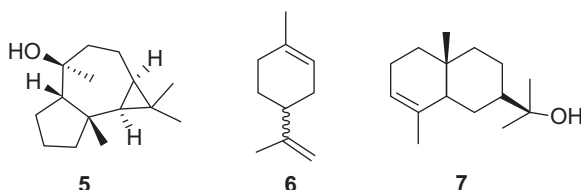


Figure 2.7 Chemical structures of viridiflorol (5), limonene (6), and α -eudesmol (7).

spectra available in instrument databases. A mixture of compounds can be separated using gas chromatography with a flame ionization detector (GC-FID), and their compounds can be identified using pure standards of the compounds of interest. With the standards analytical curve, it is possible to use the technique to quantify the chemical constituents of interest.

Studies regarding the use of gas chromatography coupled with mass spectrometry are widely reported for the separation of EO compounds from several species (de Jesus et al., 2020; Romoli et al., 2022), especially cloves, ginger, oregano, and lavender, which are internationally recognized as safe and active in food preservation (Masyita et al., 2022).

Sadekuzzaman et al. (2018) found the effectiveness of thyme and tea tree EOs against *L. monocytogenes* and *Salmonella* spp., pathogens that contaminate food, and reported the composition of these oils from the results of the analysis performed by GC-MS. The terpenes viridiflorol (5), limonene (6), and α -eudesmol (7, Fig. 2.7), main compounds of the EOs of *Psidium cattleianum* Sabine, *Cryptocarya aschersoniana* Mez., and *Ocotea*

dispersa Nees (Mez.), were identified through GC-MS by Dias et al. (2022) and were able to expressively inhibit the growth of microorganisms such as *Bacillus cereus*, *Salmonella typhimurium*, *Brochothrix thermosphacta*, and *Clostridium botulinum*.

2.3.2 Classical liquid chromatography

Conventional liquid chromatography is the most common method for separating metabolites from plant extracts and is often used as a prior purification technique for these samples. Subsequently, plant extracts are subjected to different separation methods, increasingly sophisticated, chosen according to the class of compounds present in these samples, such as triterpenes, alkaloids, or flavonoids. In addition the selected technique considers the molecular weight, stereochemistry, and polarity of the substances (López-fernández et al., 2020).

The separation is usually performed in open glass tubes (columns) under low or environmental pressure (Lanças, 2016). The compound separation in this method occurs as a function of the polarity similarity of the sample compounds with the mobile phase or the stationary phase. Considering the separation mechanisms that occur in this type of chromatography, the following are found:

- Adsorption: The stationary phase is usually silica or alumina, and the mobile phase is a nonpolar (hexane) or average polar solvent (chloroform, acetone, ethyl acetate), in which mixtures can be used. In this case the chromatography is characterized as the normal phase. For the separation of flavonoids and other phenolic compounds, reversed-phase chromatography is usually used, in which the eluent system is a solvent (or a mixture of solvents of higher polarity), such as acetonitrile and water acidified with formic acid. The stationary phase usually consists of a column of C-18 (Lanças, 2016). Selecting appropriate stationary phases might be important to ensure that different compounds can be selectively isolated. Recently, it has been reported that chemical-bonded stationary phases, such as those composed of silica bonded to phenyl groups, allow additional selectivity for the separation of natural products due to their capability to provide π – π interactions between the electron-rich phenyl ring and a solute, which modifies the retention mechanism and contributes to the separation of compounds (Li et al., 2020). Janas et al. (2016) reported the separation of 11 flavonoids on different phenyl-bonded stationary phases in reversed-phase

conditions. The phenyl-bonded silica-based stationary phase could compensate for the polar selectivity of the octadecyl-bonded silica-based stationary phase during the separation procedure.

- Partition, which is a process in column chromatography, is based on the distribution of the solute between the mobile phase and the stationary phase. Countercurrent chromatography is a method based on this separation mechanism.
- Size exclusion: The stationary phase consists of a material of controlled porosity, which retains molecules smaller than the diameter of the pores of the stationary phase, causing larger molecules to leave the column more quickly (Lanças, 2016).

Tian et al. (2018) described size exclusion chromatography using Sephedex LH-20 as an initial fractionation step of *Finnish berry* extracts, a source of catechins.

2.3.2.1 High-performance liquid chromatography

Concerning HPLC, high pressures are used to force the eluent through the columns, where stationary phases are generally reversed, using C-18, guaranteeing successful separation (Harris, 2005). This method consists of an HPLC run using a gradient mobile phase consisting of two or more reagents, typically polar organic solvents such as methanol or acetonitrile and a weak acid such as phosphoric or phosphoric acetic acid (Wojtanowski & Mroczek, 2020) (Fig. 2.8). The HPLC chromatograph uses retention time for qualitative analysis and the chromatographic peak area or ionic peak response signal for quantitative measurements (Bian et al., 2022).

The analytes are then identified and quantified (Fig. 2.9) with a UV-Vis diode array detector (DAD) through the comparison to standard solutions of the compounds probably present in the sample (Mizzi et al., 2020). This is the usual separation method of flavonoids as RP-C18 and RP-C8 columns avoid the probability of retention of these compounds, which would occur in the normal phase columns due to the high polarity of flavonoids, mainly when associated with carbohydrates (Sammani et al., 2021). Milevskaya et al. (2019) mentioned the use of HPLC in separating catechins from extracts of many plants from the Lauraceae and Hypericaceae families. Rutin (2) as well as quercetin (8) and myricetin (9) was isolated from *Dysphania ambrosioides* (L.) extracts (Zohra et al., 2019) by the same technique.

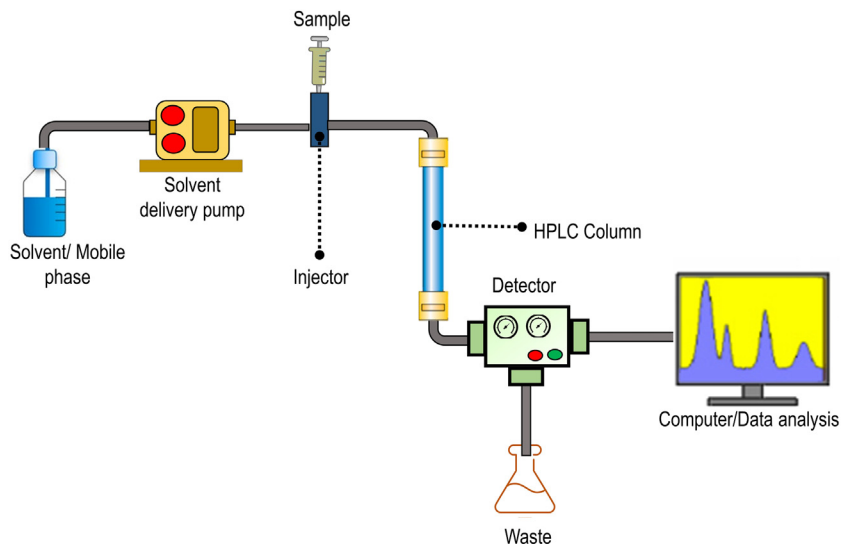


Figure 2.8 Configuration of an HPLC system.

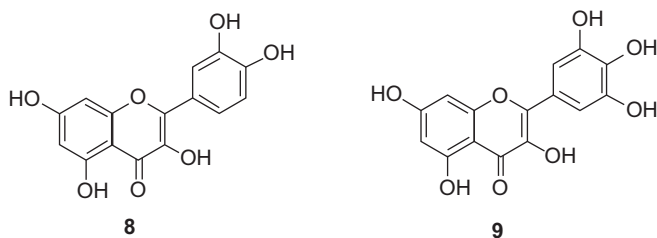


Figure 2.9 Chemical structures of quercetin (8) and myricetin (9).

2.3.2.2 Ultra-high-pressure liquid chromatography

Although HPLC is a valuable and traditional chromatographic technique to separate flavonoids, it requires a significant amount of organic solvents. UHPLC brings relevant progress working with short 10–50 mm columns filled with a 1.7–3 μm particle size compared to the 100–250 mm HPLC columns filled with 3–5 μm particles. UHPLC can significantly shorten the analytical time, maintaining efficiency and resolution power (Bian et al., 2022). The improvement in chromatography using the UHPLC system also helps minimize sample-matrix effects, such as ion suppression, that can often occur. As the optimum linear velocity has a broader range, UHPLC also allows more rapid analysis without loss of resolution (Zhang et al., 2018). UHPLC, coupled with mass spectrometry, is

gaining popularity for reliable and fast scanning capability with high resolution and accurate mass analysis (Lei et al., 2019). Examples of determination of isoflavones, flavonols, and catechins, among others, using UHPLC coupled to Ms were mentioned in the review by Bian et al. (2022), which highlighted this method's efficacy.

2.3.2.3 *Bidimensional liquid chromatography*

A chromatographic method is considered multidimensional when two or more chromatographic columns with different stationary phases are combined, implying a more efficient and faster separation. These columns, called dimensions, should not interfere with each other in the analyte separation mechanisms (Jacques & Caramão, 2009). Examples of two-dimensional chromatographic methods related to gas chromatography (Filgueiras, 2018) and thin-layer chromatography (Cortes, 2020) are reported in the literature. Considering liquid chromatography, comprehensive two-dimensional liquid chromatography (LC $\times$ LC) and HPLC-2D are described. The instrumentation used in this type of separation involves two columns connected in series, one or two high-pressure pumps, an injector, an interface or modulator, and a detector. The interface is generally a high-pressure valve, which ensures that fractions collected in the first dimension, 1D, in aliquots of predefined volumes, are continuously injected into the second column, 2D.

The critical point in two-dimensional liquid chromatography separations is transferring the elute fractions from 1D to 2D. These transfers can be carried out in two ways: online or offline. The offline mode is generally used in the pretreatment of complex samples. In this case the fractions eluted from 1D are manually collected in a flask and are reinjected later (concentrated or not) into 2D (Mogollón et al., 2014). A more efficient way of transferring the fractions to the second column is through the online technique, which has the advantage of automation, thanks to the use of pneumatic or electronic control valves, which reinject the 1D eluate directly into the 2D. The advantages of the online transfer are the analysis time. Ideally the analysis time in the second dimension should be less than or equal to the sampling rate, which corresponds to the period of transfer of fractions from the first to the second dimension (Jacques & Caramão, 2009). Several studies demonstrate how LC $\times$ LC has attracted interest in areas where samples for analysis are complex, such as metabolomics (Lv et al., 2019), pharmaceuticals (Iguiniz & Heinisch, 2017), synthetic polymers, and food (Cacciola et al., 2017), among others.

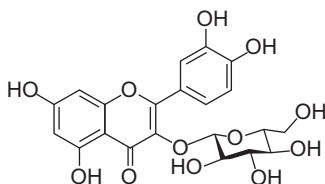
Considering bidimensional HPLC, as in the LC 2D, the separation of compounds using this technique applies two columns of two different orthogonalities. This is also referred to as LC $\times$ LC technique. Two different HPLC columns are connected in a categorization way, and the mobile phase from 1D is transferred to 2D (Jacques & Caramão, 2009). Considering chiral separations, this kind of HPLC is better at resolving complex molecules, which leads to better selectivity (Ali et al., 2021). Li et al. (2020) described the use of the two-dimensional HPLC method to carry out a high-performance separation of coumarin and flavonoid glycosides from the roots of *T. asiatica*. The isolated compounds presented high purity (Li et al., 2020).

High-speed countercurrent chromatography (HSCCC) is a separation process in which a liquid phase is retained in the spiral column by centrifugal force while a second immiscible liquid phase continuously passes through it. As it is a technique that does not require a solid support the retention of solutes is determined exclusively by their partition coefficients, which eliminates the problem of adsorption of solutes in the stationary phase (Leitão, 2005). Studies report that the greatest advantages of this separation technique, compared to preparative HPLC, are the lower amounts of mobile phase required, the nonadsorption of analytes (compounds of interest) in the stationary phase, and the complete recovery sample. However, HPLC has better resolution and a shorter analysis time than the HSCCC technique (Março et al., 2008). In addition, as it is a cheaper technique, with the use of little solvent, high sample recovery, and good separation, it is a more environmentally friendly method compared to other processes.

The HSCCC technique also allows the purification of components of complex mixtures in a single step (Sun et al., 2021), making it more attractive. Considering phenolic compounds, there are several descriptions of the successful separation of anthocyanins, proanthocyanidins, and their derivatives by this method from plant extracts of different species (Li et al., 2020).

Yang et al. (2019) highlight the high degree of purity in rutin (2, 97.3%) and isoquercitrin (10, 99.5%, Fig. 2.10) from the aerial parts of *Salvia miltiorrhiza* Bunge, proving the efficiency of this method.

In addition to gas chromatography, HSCCC can be associated with other methods and structural determination of compounds. For example, Leitão et al. (2017) reported using high-speed countercurrent chromatography combined with HPLC coupled with an ultraviolet detector coupled



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Figure 2.10 Chemical structure of isoquercitrin.

with a high-resolution mass spectrometer to detect flavonoids and phenylpropanoids from extracts of *L. origanoides*, an important Brazilian medicinal plant.

2.3.3 Supercritical fluid chromatography

SFC fills the gap between liquid chromatography and gas chromatography as the properties of the solvent are in the interface from a liquid to a solid property (Harris, 2005). The use of supercritical carbon dioxide in the mobile phase can improve chromatographic analyses in terms of resolution through improved efficiency and selectivity and reduces analysis duration because the diffusion coefficients of the solutes in supercritical fluids are higher (Lesellier & West, 2022).

SFC uses compressed carbon dioxide (CO₂), which shows the advantages of inertness, nontoxicity, nonflammability, and tunable solvent strength by varying the temperature and pressure (Harris, 2005). Furthermore the CO₂ mobile phase can be recovered to effectively reduce the environmental pollution caused by organic solvents (Bian et al., 2022).

Usually, supercritical CO₂ is mixed with a certain amount of organic solvents (such as methanol, isopropanol, hexane) added or not with bases or organic acids. This allows a better solubility of the sample in the mobile phase (Huang et al., 2018). This is noticed in the separation of flavonoids due to their polarity and acid behavior (Bian et al., 2022). Hence it is possible to use this technique in both the reversed-phase mode with nonpolar stationary phases and the normal phase mode with more polar SP, enabling the same stationary phases as in HPLC (Da Silva & Collins, 2014). As a powerful and promising technology, SFC has attracted increasing interest from researchers worldwide because it can be successfully applied to analyze terpenoids, flavonoids, and alkaloids, among other

metabolites (Huang et al., 2018). Therefore it is worth highlighting the faster analysis provided by this technique. Jiang et al. (2019) describe the efficient separation of six flavonoids, including hesperidin, isolated from *Citrus reticulata* and Pericarpium using SFC. In this study the analysis time of the same sample was evaluated using SFC and HPLC. In the first one the separation of substances was completed in 10 min, in contrast to 16 min of analysis performed by HPLC. SFC was also efficient in separating limonoids present in the EOs from 11 species of the *Citrus* genus (Zoccali et al., 2018).

In addition to the techniques for the organic compounds' separation, structural characterization is of great importance. Generally the bioactive chemical compounds of interest are separated from the matrix, which may or may not be isolated, and are characterized by a specific technique, such as EOs, which often use mass spectrometry. However, for more complex molecules, of higher molecular weights containing different functional groups, the characterization must be performed using several techniques, especially when bioactive chemical compounds are isolated for the first time. General information about the main techniques for the structural characterization of organic compounds is presented below.

2.3.4 Capillary electrophoresis

Electrophoresis is based on the differentiated migration of electrically charged species, dissolved or suspended in an electrolyte, through the application of an electric field. This technique requires a small amount of sample, aqueous buffer systems, and a reduced volume of solvent and generates little organic residue, meeting more effectively the requirements of green chemistry (Tavares, 1997). Furthermore the possibility of direct injection of complex samples, such as plant extracts or blood plasma, with no need for prior treatment, enables the systematic evaluation of several components in a single run.

For the separation of flavonoids, studies report capillary electrophoresis in two main modes of operation. One of them is the zone capillary electrophoresis, in which the capillary tube is filled with an electrolyte, usually with buffering characteristics. The separation occurs due to two strategies: maximizing the differences between the effective mobilities of solutes and minimizing the causes of zone widening (Harris, 2005).

The second electrophoretic method is micellar electrokinetic chromatography (MEKC), in which ionic surfactants, under appropriate

conditions for micelle formation, are added to the running electrolyte, providing a two-phase chromatographic system. The electrolyte represents the primary phase, which is transported electroosmotically under the action of the electric field, while the micelles represent the secondary or pseudo stationary phase, which is transported by a combination of electrophoresis and electroosmosis (Tavares, 1997). The differentiated partition of neutral solutes between these two phases is responsible for the selectivity of the separation (Wu et al., 2018). Analyses performed by capillary electrophoresis are fast, which highlights the advantage of using this technique.

Quercetin (8) and kaempferol (11) metabolites were isolated from *Hippophae rhamnoides* using capillary zone electrophoresis (Kubczak et al., 2022). Similarly, hesperidin (12), chrysin (13), epicatechin (14), and epigallocatechin gallate (15) were separated and quantified from citrus fruit juice samples and tea samples (Memon et al., 2018). These same compounds, in addition to apigenin (16) and morin (17, Fig. 2.11), which are found in wine, were also determined by capillary zone electrophoresis (Wu et al., 2018). Polak et al. (2019) described the isolation of eight isoflavones in Puerariae samples by MECK. This chromatographic process was used by Bonoli et al. (2003), who described the obtention of eight catechins and three xanthines from green tea.

2.4 Techniques for characterization of natural bioactives

The structural elucidation of organic compounds is performed by several techniques, such as UV-Visible, IR, NMR, and Ms. Each of these techniques provides helpful information in determining the molecule's structure; interpreting the spectra is similar to putting a “jigsaw” together 1.

Spectroscopy involves studying the interaction of matter with electromagnetic radiation, which can be broadly classified as absorption and emission. In absorption, electromagnetic radiation from a source is absorbed by the sample and decreases radiant energy reaching a detector. In emission, electromagnetic radiation emanates from the sample, increasing radiant power reaching a detector (Powsner et al., 2013).

Spectrometry and spectrometric methods refer to the measurement of radiation intensities using photoelectric transducers or other electronic devices (Milman, 2015).

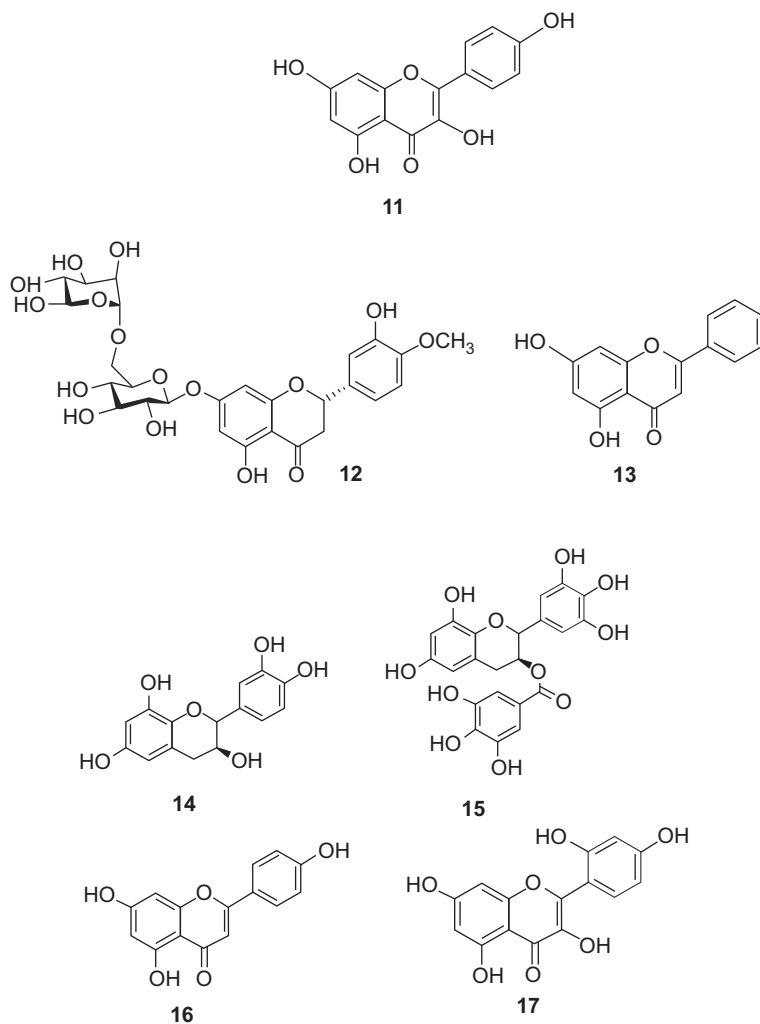


Figure 2.11 Flavonoids isolated by zone capillary electrophoresis: kaempferol (11), hesperidin (12), chrysin (13), epicatechin (14), epigallocatechin gallate (15), apigenin (16) and morin (17).

2.4.1 Principle of ultraviolet-visible spectroscopy

UV-Visible is based on measurements of the absorption of electromagnetic radiation in the visible and ultraviolet regions of the spectrum. The amount of light absorbed by the sample is measured and related to the concentration of the analyte (Mohammed, 2018).

The absorption of UV-Visible radiation occurs because the molecules have electrons that can be promoted to higher energy levels by absorbing energy. In some cases the energy required is provided by radiation with visible wavelengths, and the absorption spectrum will be in the visible region. In other cases, greater energy is required, associated with UV radiation (Passos & Saraiva, 2019).

Organic compounds that present double bonds absorb strongly in the UV region. Molecules that present conjugated single and double bonds absorb at longer wavelengths; the longer the conjugated unsaturated system, the longer the absorbed wavelengths reach the visible region. The conjugate system of a compound is called “chromophore.” The maximum absorption wavelengths of a compound make it possible to identify the chromophore it contains. Thus the UV-Visible technique is beneficial for the quantitative determination of compounds containing absorbing groups or chromophores (Rocha et al., 2018).

2.4.2 Fourier transform infrared spectroscopy

FTIR is a technique used to determine the presence of functional groups in organic compounds, such as carboxylic acids, amines, amides, aliphatic and aromatic structures, and hydroxyl groups (Nandiyanto et al., 2019).

In functional groups, there is a specific arrangement of atoms, which, when absorbing energy in the infrared region, provide rotational and vibrational movements of the molecular groups and chemical bonds of a molecule. Therefore the absorption of infrared radiation causes an increase in the amplitude of molecular vibrations. The positions of the bands in the spectrum can be presented in wave numbers using the inverse centimeter unit ($4000\text{--}400\text{ cm}^{-1}$) or in micrometers ($2.5\text{--}16\text{ }\mu\text{m}$). The FTIR technique presents many advantages, including the speed of the spectra obtention; it is nondestructible, presents high resolution, and has good reproducibility (Campanella et al., 2021).

2.4.3 Mass spectrometry

The Ms technique is used to obtain the molecular mass and structural characteristics of organic compounds by measuring their ions' mass/charge ratio. For Ms analysis the main requirement is the formation of free ions in the gas phase. After an ionization process the ionized molecule decomposes, creating ions of smaller masses that, when detected, generate the mass spectrum. There are several devices for separating ions with different

values of mass–charge (m/z) and counting in a detector. The mass spectrum of a compound is presented in the form of a graph of the number of ions versus the m/z ratio, and each molecule has a specific mass spectrum; the mass spectrum works as the “fingerprint” of a compound (Špánik & Machyňáková, 2018; Thomas, 2019).

2.4.4 Nuclear magnetic resonance spectroscopy

NMR is the most powerful tool for the structural elucidation of an organic compound. Nuclei that have spins of $+1/2$ and $-1/2$ can be studied by NMR, for example, ^1H , ^{13}C , ^{15}N , ^{19}F , and ^{31}P (Cao et al., 2021).

The spectrum obtained by ^1H NMR gives the number of signals that show us how many different types of hydrogens are present; the location of the signals shows how the hydrogen nucleus is protected or unprotected; the signal strength shows the number of hydrogens of the same type, and the splitting of the sign shows the number of hydrogens of the adjacent atoms (Andrushko & Andrushko, 2013).

To characterize organic compounds, in addition to the data obtained by analyzing the ^1H NMR spectrum the analysis of the ^{13}C NMR spectrum is of great importance to identify the carbon atoms in an organic molecule (Sharmin & Zafar, 2017). Combining IR, Ms, ^1H NMR, and ^{13}C NMR data, it is possible to elucidate the structure of organic compounds. Two-dimensional NMR can also be used, in which case a set of NMR methods provide data plotted in a space defined by two frequency axes instead of using just one (Hatzakis, 2019; Hosseinzadeh et al., 2013).

2.5 Concluding remarks: an industrial point of view

In an industrial scale, extraction yield, solvents, time, and energy consumption are relevant parameters for selecting a technology. Then, emergent extraction techniques have significant advantages compared to conventional ones from an industrial point of view. As discussed above, US, MW, PEF, supercritical extraction, and flash extraction have a higher extraction yield and consume less solvents, energy, and time. However, in some cases, the available equipment can be expensive. In this way a collaborative effort from researchers to develop affordable equipment for large-scale use is still necessary.

Regarding US, some researchers highlight the reactor design as the main challenge for reaching a scale-up of this technology. As the sound

waves become less energetic when moving away from the transducer, nonuniform cavitations regions can be formed, making the process ineffective. Therefore, continuous processes or serial reactors are suitable alternatives for industrial applications (Flores et al., 2021). In this sense, for industrial scale, a company from Germany developed a large-scale probe device for continuous process. The probe can be acquired in a power range from 500 to 16,000 W. Industrially the reactors can have a variable capacity of 5 L or L/h from 1000 L or L/h in batch or continuous processing, respectively. Equipment prices can reach up to US\$ 220,000 depending on the size of US (Vernès et al., 2019). Despite the higher cost related to the initial investment the use of US in the extraction of compounds has reduced final costs due to less time and energy consumption. Vernès et al. (2019) prove that UAE of spiruline is suitable for industrial applications considering the yield and costs.

In the same way, Solanki et al. (2019) demonstrated that MW-intensified extraction could be considered to obtain EO from *Cymbopogon winterianus* at a larger scale due to improved yield and energy efficiency. A potential industrial application to solvent-free MW extraction was presented by Filly et al. (2014), who performed EO extraction from *Rosmarinus officinalis* using a 75 L microwave reactor. A better energy efficiency and a shorter extraction time were also observed in the large-scale experiment than a conventional technique (hydrodistillation), proving the suitability of MW extraction for scaling up. On the other hand the non-conventional technique of PEF needs extensive research until it is applied on an industrial scale, regarding the optimization of parameters, increase in productivity, and equipment costs. Electric field intensity, pulse number, and frequency are the most relevant parameters for extraction yield (Naliyadhara et al., 2022). A study conducted by Tzima et al. (2021) dealt with the combination of PEF and US to extract phenolic compounds from fresh rosemary and thyme byproducts. The higher yield and low energy input indicate industrial application potential but, still, need further optimization.

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